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SECTION A — MULTIPLE CHOICE QUESTIONS (30 MARKS, 2 MARK FOR EACH)

CHOOSE THE BEST ANSWER.

(1) Which of the following best explains why semantic elements (like `<article>` and `<header>`) are preferred in HTML5 layout?

- A. They automatically add browser-specific styles.
- B. They improve screen-reader interpretation and SEO ranking.
- C. They reduce page loading time.
- D. They are easier to style with CSS than `<div>`.

(2) What will this link do when clicked?

`<a href="mailto:ray.yang@polyu.edu.hk">Email Me</a>`

- A. Opens the user's email client to compose a new message.
- B. Redirects to a web-based contact form.
- C. Submits a form to the email address.
- D. Sends an automated email directly.

(3) Which of the following CSS selectors has the highest priority?

- A. `p.intro` child
- B. `#intro`
- C. `div p`
- D. `*` Universal

CSS selector specificity determines which rule applies when multiple selectors target the same element. Browsers calculate it as a tuple (a, b, c), where:

a = number of ID selectors

b = number of class, attribute, or pseudo-class selectors

c = number of type (element) selectors or pseudo-elements

A: `p.intro` — 0 IDs, 1 class (`.intro`), 1 type (`p`) → specificity (0,1,1)

B: `#intro` — 1 ID (`#intro`), 0 classes, 0 types → specificity (1,0,0)

C: `div p` — 0 IDs, 0 classes, 2 types (`div` and `p`) → specificity (0,0,2)

**D: `*` — universal selector → specificity (0,0,0)

Comparison starts from the left: higher a wins immediately. `#intro` has a=1, while `p.intro` has a=0, so it has the highest priority. No number of classes or elements can outweigh a single ID.

(4) Given the following CSS:

`c1 { color: red; }`

`c2 { color: blue; }`

`c3 { color: green; }`

Which color will apply to `<p class="c1 c2 c3">` inside a `<div>`?

- A. Red
- B. Blue
- C. Green
- D. Cascade makes it undefined

(5) Which property does not affect the total box size calculated by the CSS box model?

- A. width
- B. border

When multiple CSS rules apply to the same element and declare the same property (color in this case), the browser uses the cascade rules to decide which one wins:

Importance and origin (e.g., !important, user/agent styles) – not applicable here.

Specificity – all three selectors (`.c1`, `.c2`, `.c3`) are single class selectors, so they have identical specificity. Order in the stylesheet (source order) – when specificity is equal, the rule that appears last in the CSS wins.

- C. padding
- D. background-color

(6) In the rule below, what does the **double colon (::)** indicate?

```
p::first-line { font-weight: bold; }
```

- C** ✓
- A. A syntax error (should be a single colon).
 - B. A pseudo-class targeting the paragraph's first line.
 - C. A pseudo-element styling the first line of text.
 - D. A vendor-specific prefix.

(7) A **complementary color scheme** consists of:

- B** ✓
- A. Three colors next to each other on the color wheel
 - B. Two colors directly opposite each other on the color wheel
 - C. One base color and two adjacent to its complement
 - D. Multiple shades of a single color

(8) Which color model is **additive**, combining light to create colors as used in screens?

- B** ✓
- A. CMY
 - B. RGB
 - C. RYB
 - D. CMYK

(9) Which of the following **pseudo-classes** styles a link that has been visited by the user?

- B** ✓
- A. a:hover This applies styles when the user hovers over the link with the mouse (or equivalent on touch devices).
 - B. a:visited
 - C. a:active This applies styles when the link is being activated (e.g., while the mouse button is held down during a click).
 - D. a:focus This applies styles when the link has keyboard focus (e.g., navigated to via Tab key and ready for a Enter).

(10) Which of the following CSS declarations will **horizontally center a block element**?

- B** ✓
- A. text-align: center; This centers inline or inline-block content inside a block element but does not center the block element itself within its parent.
 - B. margin: auto;
 - C. align: center; This is not a valid CSS property and has no effect.
 - D. float: center; This is invalid because the float property does not support a center value; it only accepts left, right, none, or related keywords.

(11) Consider the HTML fragment, Which CSS rule changes **only** the first paragraph (direct child of .content) to blue?

- B** ✓
- A. .content p { color: blue; }
 - B. .content > p { color: blue; }
 - C. .content + p { color: blue; }
 - D. .content ~ p { color: blue; }

```
<div class="content">
  <h2>Title</h2>
  <p>First paragraph</p>
  <div>
    <p>Nested paragraph</p>
  </div>
</div>
```

(12) Which rule selects items 2 and 5 only?

- A. `li:nth-child(3n-1)` 2,5
 B. `li:nth-child(3n+2)` 5,8
 C. `li:nth-child(4n+2), li:nth-child(5n)`
 D. `li:nth-child(3n-1):not(:nth-child(2n))`

```
<ol>
  <li>One</li>
  <li>Two</li>
  <li>Three</li>
  <li>Four</li>
  <li>Five</li>
</ol>
```

(13) If a `<div>` has the following styles, what is the **total width** the element occupies on the page?

```
div {
  width: 300px;
  padding: 20px;
  border: 10px solid black;
  margin: 0;
}
```

- A. 300px
 B. 330px
 C. 350px
 D. 360px

300 + 20 + 10

Total occupied width = width + left padding + right padding + left border + right border

Count Two times of Padding & Border

(14) When two vertical margins meet, what happens?

- A. The margins add up.
 B. The margins overlap, and only the larger margin is used.
 C. Both margins remain and are applied separately.
 D. The parent's margin overrides the child's margin.

(15) Given the following code:

```
<input type="number" min="10" max="20" value="5">
```

What will happen when this input field is displayed in a modern browser?

- A. The input will display 5 and allow values from 5 to 20. *Incorrect — it does not allow values from 5 to 20; the range is still 10–20. Values below 10 are invalid.*
 B. The input will display 5, but the form will fail validation if submitted.
 C. The input will display 10 as the default value. *Incorrect — the browser does not automatically change the displayed value to the minimum (10). It keeps showing 5.*
 D. The input will be invalid and not render.

Incorrect — the input is perfectly valid HTML and will render normally. Being out-of-range does not prevent rendering.

Q10:<video controls>
<source src="promo.mp4" type="video/mp4">
</video>

SECTION B — SHORT QUESTIONS (40 MARKS, 4 MARKS FOR EACH)

NOTE: ANSWER LESS THAN 10 SENTENCES FOR EACH QUESTION.

Q1. Explain the difference between **inline**, **embedded**, and **external** CSS.

Q2. Write CSS code that makes all `<h2>` elements display uppercase text, apply 10px letter spacing, and center the text horizontally.

Q3. Write a CSS rule that applies a background image (`pattern.png`) which repeats only horizontally and stays fixed when the page scrolls.

Q4. Please explain why "Times New Roman" is enclosed in quotation marks in the following CSS rule:

```
p { font-family: "Times New Roman", Sans-serif, serif; }
```

Cascading" (瀑布式/级联) 描述事物像瀑布一样层层叠叠、逐级向下流淌或影响的现象

Q5. In one sentence, explain the purpose of the `alt` attribute in an HTML `<img>` tag.

Q6. How do you understand the term "cascading" in *Cascading Style Sheets (CSS)*?

Q7. What is the main difference between the **border** and the **outline** properties in CSS?

Q8. Write a CSS rule that creates a box with a width of 300px, a height of 200px, a solid red border 5px thick, and 20px of padding inside the box.

Q9. Describe the main difference between **block-level elements** and **inline elements** in HTML. Provide one example of each.

Q10. Write the HTML code to embed a video file named **promo.mp4** with playback controls enabled.

Q1:Inline CSS is written directly in an HTML element's style attribute, affecting only that element. Embedded CSS is placed inside a `<style>` block in the HTML document's `<head>`, affecting the whole page. External CSS is stored in a separate .css file linked via `<link>`, allowing it to be reused across multiple pages.

Q2: reuse across multiple pages

```
h2 {  
  text-transform: uppercase;  
  letter-spacing: 10px;  
  text-align: center;  
}
```

Q3

```
body {  
  background-image: url("pattern.png");  
  background-repeat: repeat-x;  
  background-attachment: fixed;  
}
```

Q8

```
.box {  
  width: 300px;  
  height: 200px;  
  border: 5px solid red;  
  padding: 20px;  
}
```

Q4:"Times New Roman" is enclosed in quotation marks because it contains spaces; multiword font names must be quoted in CSS to be recognized as a single font family name

Q5:The alt attribute provides alternative text for an image if it cannot be displayed, used by screen readers and for SEO.

Q6: "Cascading" refers to the way CSS resolves conflicts when multiple rules apply to the same element, based on source order, specificity, and importance

Q7:The border is part of the box model, affecting layout and dimensions; the outline is drawn outside the border, does not take up space, and is often used for focus indicators.

Q9:Block-level elements take the full width of their container and start on a new line (e.g., `<div>`), while inline elements take only necessary width and stay on the same line of text (e.g., `<span>`).

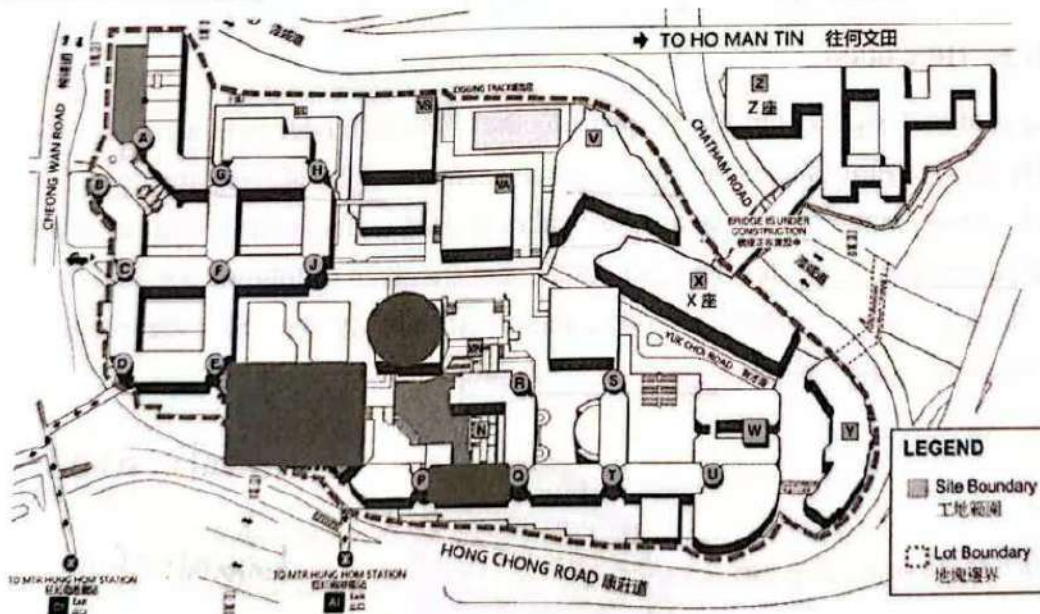
SECTION C — LONG QUESTIONS (30 MARKS)

QUESTION 1: (10 MARKS)

Given the PolyU campus map image, write an HTML program to create an image-based map. When a user clicks on the PQ Building, the page should open the COMP Department website at <http://comp.polyu.edu.hk>; when clicking the E Building, it should open the Library website at <http://library.polyu.edu.hk>; and when clicking the M Building, it should open the HRO Department website at <http://hro.polyu.edu.hk>. The buildings are highlighted in red on the image, and their geometric properties are as follows.

- (1) **COMP:** The PQ Building is approximated as a rectangle whose top-left corner is located at coordinates (350, 300) with a width of 100 pixels and a height of 50 pixels.
- (2) **Library:** The E Building is also approximated as a rectangle whose top-left corner is at coordinates (170, 260) with a width of 200 pixels and a height of 100 pixels.
- (3) **HRO:** The M Building is approximated as a circle whose center is at coordinates (325, 255) with a radius of 50 pixels.

You are required to write the complete HTML code that creates this interactive image map so that each specified region correctly links to its corresponding webpage when clicked.



```

<map name="polyuMap">
  <area
    shape="rect" coords="350,300,450,350"
    href="http://comp.polyu.edu.hk"
  />
  <area
    shape="rect" coords="170,260,370,360"
    href="http://library.polyu.edu.hk"
  />
  <area
    shape="circle" coords="325,255,50"
    href="http://hro.polyu.edu.hk"
  />
</map>
```

QUESTION 2: (10 MARKS)

Consider the following HTML document with the embedded CSS code. What will be the **color of the heading text (h1)** displayed on the web page? Explain why this color is applied, referring to how the browser interprets and applies the given CSS rules.

The selector `:not(p)` means "select every element that is NOT a `<p>` element." So, this rule applies `color: red;` to all elements except `<p>`

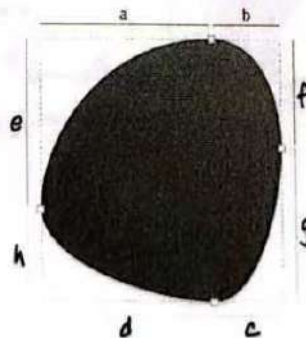
```
<!DOCTYPE html>
<html>
<head>
<style>
:~not(p) {
  color: red;
}
</style>
</head>
<body>|

<h1>This is the heading 1</h1>
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

</body>
</html>
```

QUESTION 3: (10 MARKS)

You are required to design a blob-like shape using the CSS box model and the `border-radius` property. The figure below shows the target shape, where the eight labeled parameters (a, b, c, d, e, f, g, and h) represent the horizontal and vertical radii used to define the corners of an element's border. The percentage values for the eight parameters are given as follows: a = 70%, b = 30%, c = 30%, d = 70%, e = 60%, f = 40%, g = 60%, and h = 40%. Write the CSS code required to create this shape by using a single border-radius property.



vertical: abcd

horizontal: efgh

```
.blob{
  width: 280px; height: 280px;
  background: linear-gradient(135deg, #ff4da6, #5b00ff);
  border-radius: ;
}
```

border-radius: 70% 30% 30% 70% ,
60% 40% 60% 40%;

END